

MARS Practical

STA5090Z: Topics in Advanced Regression

August 2026

Question 1

Explore R's `earth()` function (in package `earth`) for MARS, using the Spam data (see link below). MARS is used as a trademark by a software company, so open-source packages for MARS are often called 'earth'.

Look at:

- **arguments:** `penalty`, `degree`, `nk`, `thresh`, `pmethod`, `nprune`.
- **returned object:** `rss`, `rsq`, `gcv`, `grsq`, `bx`

Compare to a GAM model for the Spam data.

Spam Data: The last column denotes whether the e-mail was considered spam (1) or not (0), i.e., unsolicited commercial e-mail. Most of the attributes indicate whether a particular word or character was frequently occurring in the e-mail. Most of these count the frequency of a particular word or type of character (one of these is George, the email receiver's first name). Attributes 55-57 measure the length of sequences of consecutive capital letters.

Spam data available at: <https://archive.ics.uci.edu/ml/datasets/spambase>

Question 2

Write a function to construct the basis for a MARS model, given a single predictor variable X . Illustrate these functions.

Question 3

Simulate data from the following model: $N = 100$ observations, predictors $X_1, X_2, \dots, X_p$ and errors ϵ have independent standard normal distributions.

Scenario 1:

$$p = 2$$

$$Y = (X_1 - 1)_+ + (X_1 - 1)_+ \cdot (X_2 - 0.8)_+ + 0.12 \cdot \epsilon$$

a.) Plot the surface (Y as a function of X_1 and X_2), without error.

b.) Fit a MARS model to the data (with error). Obtain R^2 . (Hastie et al. obtain a mean R^2 of 0.97, 5 simulations.)

Scenario 2:

Scenario 1 but $p = 20$, i.e. there are an additional 18 inputs unrelated to the response.

ESL mean $R^2 = 0.96$.

Scenario 3:

This has the structure of a neural network:

$$\ell_1 = X_1 + X_2 + X_3 + X_4 + X_5$$

$$\ell_2 = X_6 - X_7 + X_8 - X_9 + X_{10}$$

$$\sigma(t) = 1/(1 + e^{-t})$$

$$Y = \sigma(\ell_1) + \sigma(\ell_2) + 0.12 \cdot \epsilon$$

ESL mean $R^2 = 0.79$.

Explore

1. Kartal Koc, E. and Bozdogan, H. (2015) Model selection in multivariate adaptive regression splines (MARS) using information complexity as the fitness function, Machine Learning, 101(1–3), pp. 35–58. <https://doi.org/10.1007/s10994-014-5440-5>
2. Do a simulation study to investigate how strongly model selection should penalize the parameters.